



Society of Petroleum Engineers

SPE-227100-MS

Downhole Leak Detection Using High Frequency DAS and DTS Signals from Fiber Optics Cables

T. Zharnikov, Aramco Innovations, Moscow, Russian Federation; D. Alekseev, BMK-Service, Moscow, Russian Federation; S. Aloqili, Saudi Aramco, Dhahran, Saudi Arabia; M. Zelenyy, BMK-Service, Moscow, Russian Federation; S. Jacob and A. Tammar, Saudi Aramco, Dhahran, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227100-MS](https://doi.org/10.2118/227100-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The aim of this work is to assess the feasibility of detecting well fluid leaks using the high-frequency part of the acoustic field in Distributed Acoustic Sensing (DAS) signals. Particularly, this advance signal analysis is carried out as part of permanent monitoring using distributed fiber optic sensing (DFO). A comprehensive approach combining numerical simulations and analytical modeling was employed to analyze the high-frequency (1-10 kHz) component of the leak-generated acoustic field. Our approach overcomes the complexities of simulating the entire wavefield spectrum by decoupling the low-frequency and high-frequency components, thus enabling a more focused investigation of the high-frequency component. A custom-built DAS model based on the previous studies ([Liokumovich et al, 2015](#); [Alekseev et al, 2020](#)) was used to simulate the high-frequency measurement process, considering the effects of spatial averaging and filtering introduced by the wellbore and surrounding formations. Spectra of the simulated acoustic field and DAS response are calculated and analyzed to identify features that may indicate leakage.

Our results demonstrate that the high-frequency component of the DAS signal can provide valuable insights into leak characteristics, including location, severity, and type. However, accurate interpretation requires careful consideration of the effects of well and DAS measurement systems on the acoustic noise. Frequency bands with high noise signals can be identified by analyzing the well system, including wellbore geometry and material, and fluids and formation elastic properties. Additionally, the DAS measurement system introduces spatial averaging of the signal, leading to smoothed and smeared signals over depth. By understanding these relationships, we have established guidelines for optimizing leak detection strategies using DAS. Notably, our findings highlight the importance of selecting optimal frequency bands and considering the impact of spatial averaging on signal quality.

This research contributes significantly to the development of advanced leak detection methods using DAS technology. By emphasizing the need to consider both well and DAS measurement system effects, our work provides new perspectives on interpreting high-frequency DAS signals. These insights enable more effective optimization of leak detection strategies, ultimately enhancing the efficiency and reliability of well integrity monitoring.

Introduction

Leakage detection in oil and gas pipelines is necessary to facilitate safe operation, minimize environmental impact, and avoid costly repairs. Distributed Temperature Sensing (DTS) and Distributed Acoustic Sensing (DAS) are the two prominent technologies utilized for leak detection ([Hveding et al, 2018](#); [Farhadiroushan et al, 2019](#)). Both utilize fiber optic cables but implement distinct principles to identify anomalies that are indicative of leaks. Distributed Temperature Sensing (DTS) is based on the utilization of fiber optic cables laid in the boreholes to measure temperature variations along the borehole. Temperature deviation occurs when fluid leaks out of the pipe, producing cooling/heating effects in the vicinity of the leak location. Through continuous temperature profile measurement, DTS detects deviations indicating potential leaks. DTS relies on Raman scattering, which is due to the interaction of laser pulses with the molecular vibrations. After the interaction, the scattered photons carry the information about the cable temperature in the amplitudes of the frequency shifted lines. The shift is caused by inelastic scattering. The amplitude of Raman scattering is governed by the fiber temperature at the measurement point. By analyzing the backscattered signals, precise temperature profiles in the borehole can be obtained.

For oil and gas applications, fiber optic cables are typically deployed inside the borehole. They can be cemented behind casing, clamped on the tubing, or put into the well in the process of intervention job. Because fluids flow through pipes, any changes in thermal conditions brought about by leakage manifest as localized temperature changes via DTS analysis. Heat from surrounding rock formations can also influence temperature measurements indirectly, yielding complementary data about formation around the borehole.

Compared to conventional monitoring techniques with pressure sensors or acoustic emission detectors alone, DTS offers the following advantages: Continuous spatial coverage can detect small issues before they escalate into catastrophic failures. High sensitivity offers early warning systems that can prevent such undesirable events like large leakages, gas breakthrough, aquifer contamination. Spatial resolution allows the determination of exact leak positions efficiently. Real-time response enables timely intervention measures minimizing environmental risks and economic losses associated with downtime repairs. Scalability makes it suitable for the wells of different length, including extended reach wells without a compromise on accuracy. Several studies exist that disclose how DTS assists in enhancing safety standards while reducing operational costs involved in maintenance activities. For instance, it is used for monitoring leak detection, estimating injectivity profile during injection jobs, inflow profiles to optimize the production. These instances demonstrate the versatility afforded by incorporating emerging technologies into traditional engineering processes with the aim of optimizing performance. It is directly linked to risk mitigation approaches employed throughout the entire lifecycle of the well. Exploitation of distributed temperature sensing methods is a promising move towards safer operations in hostile environments with high pressures, corrosive chemicals, harsh climatic conditions, and inaccessibility. With ongoing development expected both in terms of hardware capabilities as well as computational methodologies, future models promise to provide greater accuracy with lower false alarm rates ultimately leading to greater overall system resilience to unforeseen disruptions.

Regarding DAS, it utilizes the fiber optic cables to identify vibrations rather than temperature. Sound waves propagating through liquids/gases leaking from leaks generate unique signatures, which are detected by sensitive distributed acoustic sensing integrated into the cable. Interrogators interpret these patterns to infer leak characteristics. Acoustic disturbances are comprised of sound created by turbulence, leakage, valves, gas bubbles and other events, providing context beyond pure thermal signatures. It was shown experimentally that high frequency signals can be registered (tens of kHz) ([Tertyshnikov et al, 2022](#)). Such signals can be produced by the leaks. The best results are obtained when fiber optics is placed in close enough proximity to allow detection of small vibrational effects of fluids leaking. The Rayleigh scattered pulses are produced by the backscattering on fiber inhomogeneities and measured by a detector. Sophisticated filtering is applied to isolate features that are tied directly to leak occurrences. To precisely estimate leak positions,

correlations between received noises are computed. The advantage of using DAS comes from the fact that it is robustness against environmental conditions since acoustic signals do not depend on temperature variations.

By combining the complementary benefits of DTS and DAS the reliability of leak detection is significantly enhanced (McKinley et al, 1973; Mutovkin et al, 2020; Miklashevsky et al, 2020). DTS excels at picking up slow leaks dominated by thermodynamic processes, while DAS is more suitable for rapid transition processes that generate big mechanical disturbance. The dual-sensor approach cross-validates the results minimizing false positives in solutions based on one of these technologies alone. Despite advances, there are still some challenges with DFO sensing. They include the signal interpretation and interference (e.g., noise discrimination from seismic activity), the cost associated with the fiber optics installation and use, the challenges of calibration processes necessary for continued operations.

Theory and method

The fiber optic signal is the sum of the individual backscattering events:

$$E(t, \tau) = \sum_{N(t)}^{N(t)+M} a_i e^{i\omega t} e^{ikx_i} \quad (1)$$

Since the detector reacts to the light intensity, it registers the quantity

$$I(t, \tau) = E(t, \tau) E^*(t, \tau) \quad (2)$$

One of DAS interrogation schemes is heterodyne detection (Hartog, 2017; Shatalin et al., 2021). In this scheme, frequency modulated backscattered signal is mixed with the reference frequency light output. Beat frequency signal is detected and extracted. Typical processing involves phase extraction from the raw signal, for example, by using Fourier transform. Then the phase is unwrapped to get continuous phase. Finally, the result is differentiated over the gauge length to get the signal, which is proportional to strain.

The strain is due to the acoustic perturbation in the borehole. The turbulent noise from the leak is one of the sources of acoustic signal in the borehole. The elastic wave propagation can be described by the acoustic equation in the fluid and the elastic wave equation in solids as follows:

$$\rho \frac{\partial^2 \vec{u}}{\partial t^2} - \nabla \cdot \tau = 0 \quad (3)$$

$$\frac{1}{\rho c^2} \frac{\partial^2 p}{\partial t^2} + \nabla \cdot \left(-\frac{1}{\rho} (\nabla p) \right) = Q \quad (4)$$

Here p is the total pressure, ρ denotes material density, c is the speed of sound, Q is the source intensity, $\vec{u}$ is displacement vector and τ is the stress tensor. Knowing the source spectrum $S(f)$ one can compute the fiber optics strain. This strain can be input into the model of fiber optics measurement to compute DAS response. By combining the two steps, DAS leak measurement can be simulated and analyzed.

The elastic field in the borehole can be modeled using finite element approach. Because typical model size is from several to tens of meters along the borehole and of similar size in radial direction, direct 3d modeling using standard software takes long time to compute. This is the case for the high frequency (1-10 kHz) signals, which we are interested in. To overcome this complication, we resorted to axisymmetric 2d modeling. DAS signal can be computed directly using Eqs. (1) and (2).

Often, the leak is accompanied by the temperature changes. The thermal signature depends on the type of fluid, leak location, flow rate, and well conditions. It can be quantified and interpreted to diagnose the type and cause of the leak. The effects can vary, but generally gas leaks lead to cooling due to Joule-Thomson effect, fluid leaks can be accompanied by adiabatic expansion (minor cooling is possible) and warming up due to viscous energy dissipation. In addition, fluid mixing plays significant role in determining the resulting temperature change. The amount of the temperature effect varies depending on the flow rate and

conditions and ranges from tenth of degree C to 10-20°C. Larger size and flow rate leaks produce larger temperature variations. One of the strategies is to compare the DTS profile against the baseline. Advanced analysis involves transient measurements. One of the methods to quantify the interpretation is to combine the measurement and analysis with the flow and temperature simulations.

Leak detection using high frequency

The following borehole model was investigated. The 60-meter vertical section of a cased borehole was assumed. It is shown schematically in Figure 1. Typical well completion diameters were considered. The radius of the borehole was taken to be approximately 6 inches, whereas steel casing has a smaller radius of approximately 4 inches. Cemented well is assumed. Tubing with the radius of approximately 2 inch was placed in the wellbore. The properties of the materials are as follows. The speed of sound in the water is 1480 m/s, which is close to that of brine or water-based drilling mud. The compressional and shear wave speeds in steel casing and tubing are 6270 m/sec and 3203 m/sec, respectively. The steel density was assumed to be 8 gcc. These values are realistic for normal steel grades utilized in well construction. The compressional and shear wave speeds of an average strength cement grout are about 5971 m/sec and 3612 m/sec, respectively. The density is 2.3 gcc. Typical compressional and shear velocities of a moderately consolidated sandstone formation, often encountered in hydrocarbon reservoirs, are approximately 3905 m/sec and 2540 m/sec, respectively. The density was taken to be 2.125 gcc. The leak is modeled as follows. A tubing leak is simulated as a monopole point source with a frequency spectrum that approximates the one measured by McKinley et al. (1973) (see Figure 1). To simplify calculations, the frequency leak signal band limit is set at 100–10000 Hz covering the most energetic acoustic frequency band for such a leak.

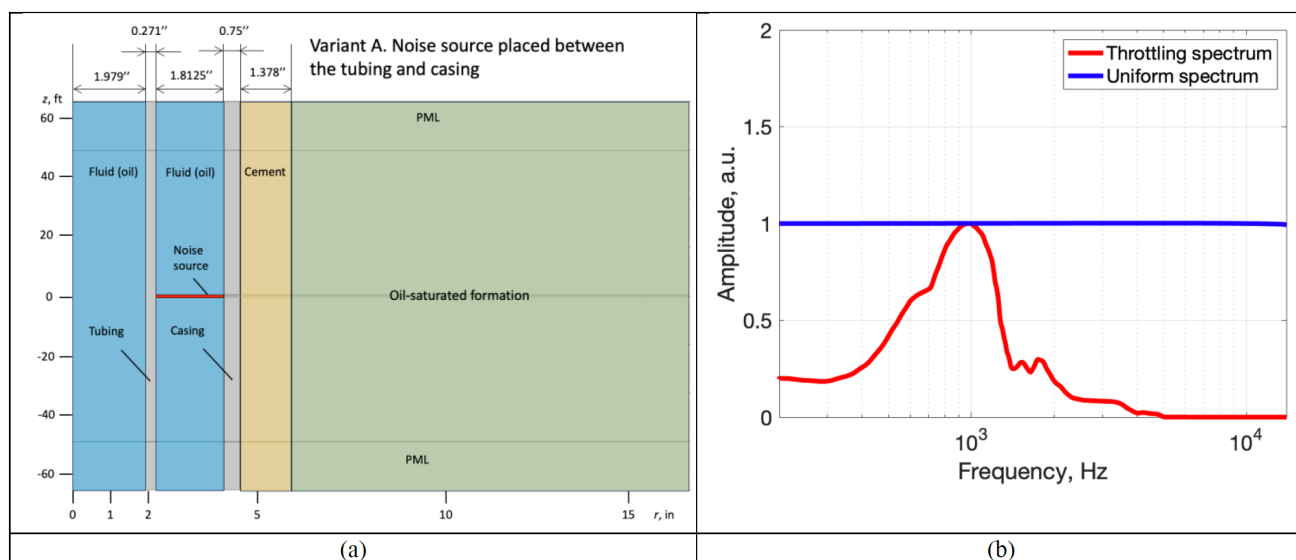


Figure 1—Sketch of the wellbore model (left) and source spectrum (right).

DAS response is simulated taking into account the following assumptions. Short, 10-nanosecond, optical pulse is utilized to allow high spatial resolution measurement of strain. The gauge length value was taken to be 1 m. Heterodyne measurement with 70 MHz shift was considered, which not unusual for ϕ -OTDR systems. A high-frequency filter is employed to process the raw unfiltered DAS data and eliminates noise and enhances detection of strain fluctuations as a result of leaks. The numerical model is axially symmetric to simplify and to speed up the computations. This is reasonable given the cylindrical geometry of the borehole and expected uniform pattern of radiation leak source. Frequencies less than 30 Hz and more than 1000 Hz are filtered out based on typical operational frequency of DAS systems and acoustic energy attenuation by

leaks in this range. This setup provides a test platform for field measurement of DAS sensitivity to detect and quantify tubing leaks under simulated downhole conditions. Other parametric studies can explore cement quality variations, formation heterogeneity, and other spectra of leaks.

The waterfall plots for the simulated DAS response and the leak noise of the flat spectrum in this example are given in Figure 2a. Leak noise behavior is consistent with the findings of Mutovkin et al. (2020). It has two main characteristics as follows. Increased amplitude closer to the source. There is significant signal amplification closer to the leak source, as expected of a monopole sound source in a fluid-filled borehole. The second feature is the appearance of resonance frequencies. DAS response has the tendency to have strong spatial localization close to the leak source. It is because DAS is picking up mostly the near-field strain variations resulting from the leak and not the tube waves. These outcomes contribute to the explanation of empirical results of Tertyshnikov et al. (2022), where they demonstrated efficient high-frequency leak localization in DAS data using no waveform stacking. The study explains that DAS is naturally sensitive to near-field strain disturbances, and consequently, it is ideally designed for leak position detection without requiring much signal processing. DAS high spatial resolution (especially with shorter gauge length) facilitates direct observation of leak-induced strain anomalies, whereas normal acoustics are more plagued by propagating tube waves.

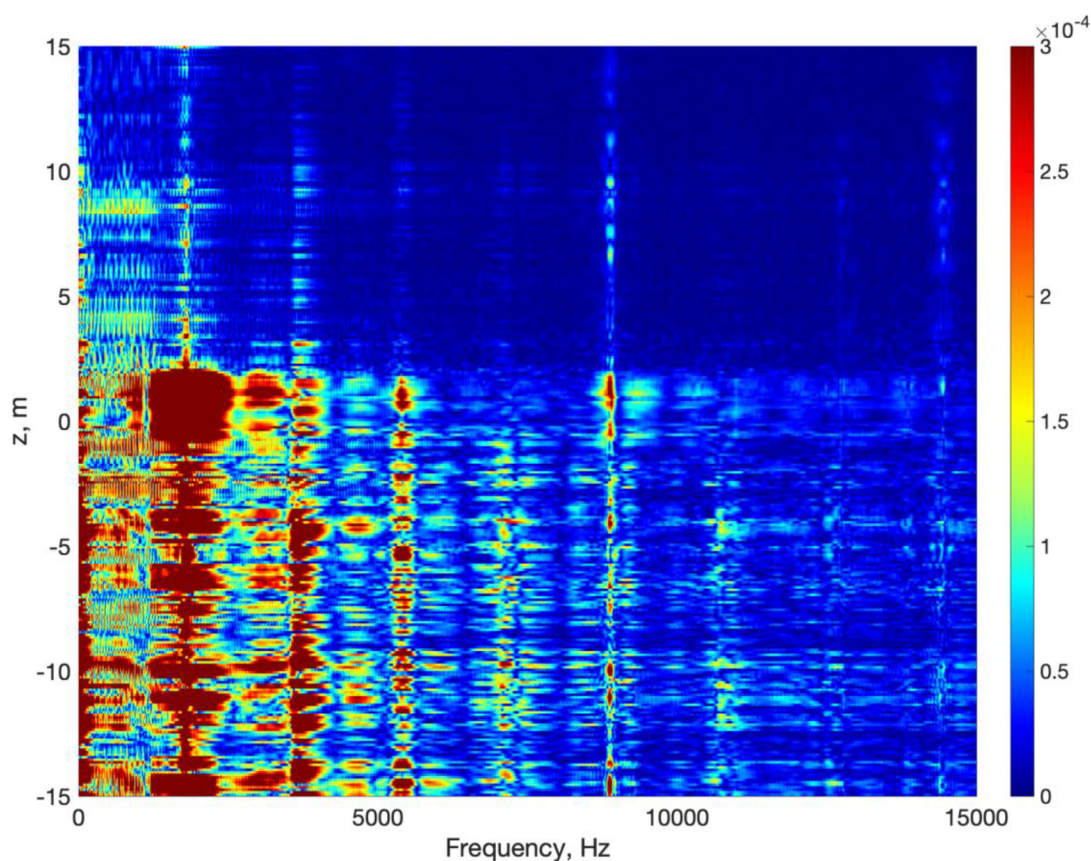


Figure 2—Spectrogram (depth is used instead of time) of modelled DAS response to acoustic field in borehole assuming broadband leak noise.

The results prove the better localization capability of DAS for high-frequency noise sources, e.g., tubing leaks, compared to common acoustic logging techniques. Future work can include the study of the gauge length effect on localization accuracy, the impact of formation properties on altering the DAS response, the field validation of such modeling results for various well conditions.

Simulated data for a broadband leak source are presented in Figure 2 (spectrograms). Compared to the single-frequency case, the spectrograms display higher levels of noise over multiple frequency components, making it more difficult to detect leaks in the time-frequency domain. Two important conclusions regarding depth accuracy are as follows. There is minimum depth shift between measurements at different gauge lengths. These findings suggest that while DAS is able to effectively identify leak presence, precise depth determination requires careful consideration of gauge length selection, potential application of multi-gauge analysis techniques and further processing to compensate for measurement biases. Multi-band analysis (FBE method) is beneficial in verifying leak detection across various frequency bands and reducing spurious responses in noisy environments. Future studies must aim at techniques of depth calibration for improving the precision of localization and adaptive gauge length selection methods.

This research verifies that despite the fact that broadband leak sources involve greater complexity compared to single-frequency situations, sufficient processing techniques and gauge length enable effective leak detection and location. The FBE method is particularly efficient in separating leaks from external noise in different frequency bands. However, the depth variations recorded with these experiments emphasize the importance of complex interpretation techniques to ensure accurate positioning of the leak in actual operating conditions.

Conclusions

We introduce an innovative fully coupled modeling strategy that honors, simultaneously, acoustic wave propagation in boreholes and Fundamental principles of distributed fiber optic sensing. This framework elucidates the fundamental distinctions between the use of conventional hydrophone measurements and existing DAS technology for leak detection. DAS response exhibits good spatial localization near the leak source. Such localization is due to the transfer function features of the tool. This study guides choice of optimal gauge lengths and helps in selection of appropriate spatial sampling intervals. By bridging the gap between borehole acoustics and the principles of fiber-optic sensing, this work provides a physical basis for field-measured data and can enhance reliable well integrity monitoring solutions.

References

1. Hveding, F. and Bukhamsin, A., 2018, "Distributed fiber optic sensing-a technology review for upstream oil and gas applications", SPE-192323-MS
2. Farhadiroushan, M., Parker, T., Shatalin, S., Gillies, A., Chen, Z., Clarke, A., Naldrett, G., 2019, "Advanced geophysical measurement methods using engineered fiber optic acoustic sensor", In the 81st EAGE Conference and Exhibition.
3. McKinley, R., Bower, F., and Rumble, R., 1973, "The structure and interpretation of noise from behind cemented casing", SPE-3999-PA.
4. Mutovkin, N., Mikhailov, D., and Sofronov, I., 2020, "Modeling acoustic fields induced by the flow noise in the near-wellbore zone", *Mathematical Models and Computer Simulations*, **12**(1), p. 70–76
5. Miklashevskiy, D., Shako, V., Borodin, I., Wilson, C., Kortukov, D., Tarelko, N., and Zozulya, O. 2020, "Approach for wellbore production monitoring using distributed acoustic noise measurements", International Petroleum Technology Conference, IPTC-20125-Abstract
6. Tertyshnikov, K., Lebedev, M. and Pevzner, R., 2022, "High-frequency acoustic data acquired with DAS: field trial at the Curtin training well", In the 2nd EAGE Workshop on Fiber Optic Sensing for Energy Applications in Asia Pacific, pages 1–4. EAGE

7. Liokumovich, L., Ushakov, N., Kotov, O., Bisyarin, M. and Hartog, A., 2015, "Fundamentals of optical fiber sensing schemes based on coherent optical time domain reflectometry: signal model under static fiber conditions", *Journal of Lightwave Technology*, JLT-17555-2015
8. Alekseev, A., Gorshkov, B., Potapov, V., Taranov, M. and Simikin, D., 2020, "Dual-pulse phase-OTDR response to propagating longitudinal disturbance", *Laser Physics*, **30**, p. 035107
9. Hartog, A., 2017, "An introduction to distributed optical fibre sensors", CRC Press
10. Shatalin, S., Parker, T. and Farhadiroushan, M., 2021, "High definition seismic and microseismic data acquisition using distributed and engineered fiber optic acoustic sensors", *In Distributed Acoustic Sensing in Geophysics, AGU: Methods and Applications*, p. 3–32